IE410 Project Part B — Report

Simulation of a Theo Jansen Walking Mechanism for Gait Generation

Group 18 — DAU, Winter 2026

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1. Introduction

The Theo Jansen mechanism is a single degree-of-freedom planar linkage that converts continuous rotary input into a complex, leg-like end-effector trajectory. Unlike conventional robotic legs that require multiple actuators and joint controllers, this mechanism uses only one crank motor to produce a smooth, gait-resembling foot path. This project simulates a modified Jansen-type gait trainer based on the 12-link, eight-bar topology described in Shin, Deshpande, and Sulzer (*JMR 2018*), with the goal of reproducing a human ankle-like trajectory and studying the effect of adjustable link dimensions on the resulting gait shape.

2. The Modeled Linkage

2.1 Topology

The mechanism follows a 12-link, eight-bar closed-loop topology. It has a single degree of freedom — the crank angle θ_1 — which drives the entire assembly. All other joint angles are determined by the closure of five vector loop equations, solved numerically at each crank position.

The twelve links are connected through eight joints (pt0 through pt6, and ptE), with the following connectivity:

Link	Connection
L1	pt0 → pt1 (crank)
L2	pt1 → pt2
L3	pt3 → pt2
L4	pt0 → pt3 (ground link)
L5	pt2 → pt4
L6	pt3 → pt4
L7	pt1 → pt5
L8	pt3 → pt5
L9	pt5 → pt6
L10	pt4 → pt6
L11	pt5 → ptE
L12	ptE → pt6

pt0 and pt3 are fixed ground pivots. L4 (the ground link) is fixed at angle $\theta_4 = 0$.

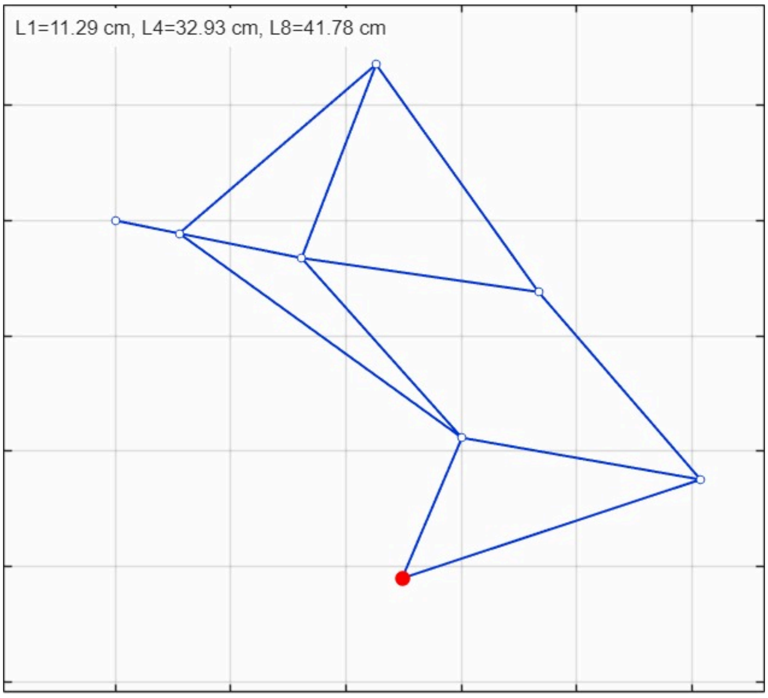

Figure 1: One-frame snapshot of the 12-link modified Jansen mechanism assembly. Blue lines represent the twelve links, hollow circles are the intermediate joints, and the red filled circle marks the foot endpoint (ptE). Link dimensions: L1 = 11.29 cm, L4 = 32.93 cm, L8 = 41.78 cm.

2.2 Link Dimensions

The nominal link lengths used in simulation, taken from the paper's appendix with the adjustable values L1, L4, and L8 substituted:

Link	L1	L2	L3	L4	L5	L6	L7	L8	L9	L10	L11	L12
Length (cm)	11.29	45.0	36.0	32.93	48.5	41.5	60.5	41.78	42.0	43.0	26.5	54.5

L1, L4, and L8 are the three adjustable links identified in the reference paper as the primary parameters for tuning gait shape.

2.3 Simulation Parameters

- **Crank speed (ω):** π rad/s (one full cycle in 2 seconds)
- **Angular resolution:** 361 steps per cycle (1° per step)
- **Frame rotation:** −11.4° applied to align mechanism frame with gait/world frame
- **Solver:** MATLAB `fsolve` (Optimization Toolbox), with a damped Newton fallback for continuity

3. The Chosen Endpoint

The tracked point is **ptE** — the foot or ankle endpoint of the linkage, located at the tip of links L11 and L12. This is the point that, in a physical walking device, would contact the ground during the stance phase and lift off during the swing phase. Its Cartesian coordinates (x, y) are recorded for every crank angle over one complete revolution, producing the closed-loop foot trajectory.

4. Generated Gait Trajectory

4.1 Trajectory Shape

Over one full crank cycle, ptE traces a smooth, closed, ellipse-like curve in the x-y plane. The trajectory has a characteristic asymmetry — the lower portion (stance phase analog) is relatively flat and extended horizontally, while the upper portion (swing phase analog) arcs upward more sharply. This shape qualitatively matches a natural ankle trajectory during level walking.

Key trajectory metrics from the simulation:

- **Horizontal span (stride length):** ~50 cm
- **Vertical span (step height):** ~12 cm
- **Enclosed loop area:** computed via `polyarea` on the ptE coordinates
- **Maximum loop closure residual:** < 10^{-6} cm, confirming reliable numerical solution

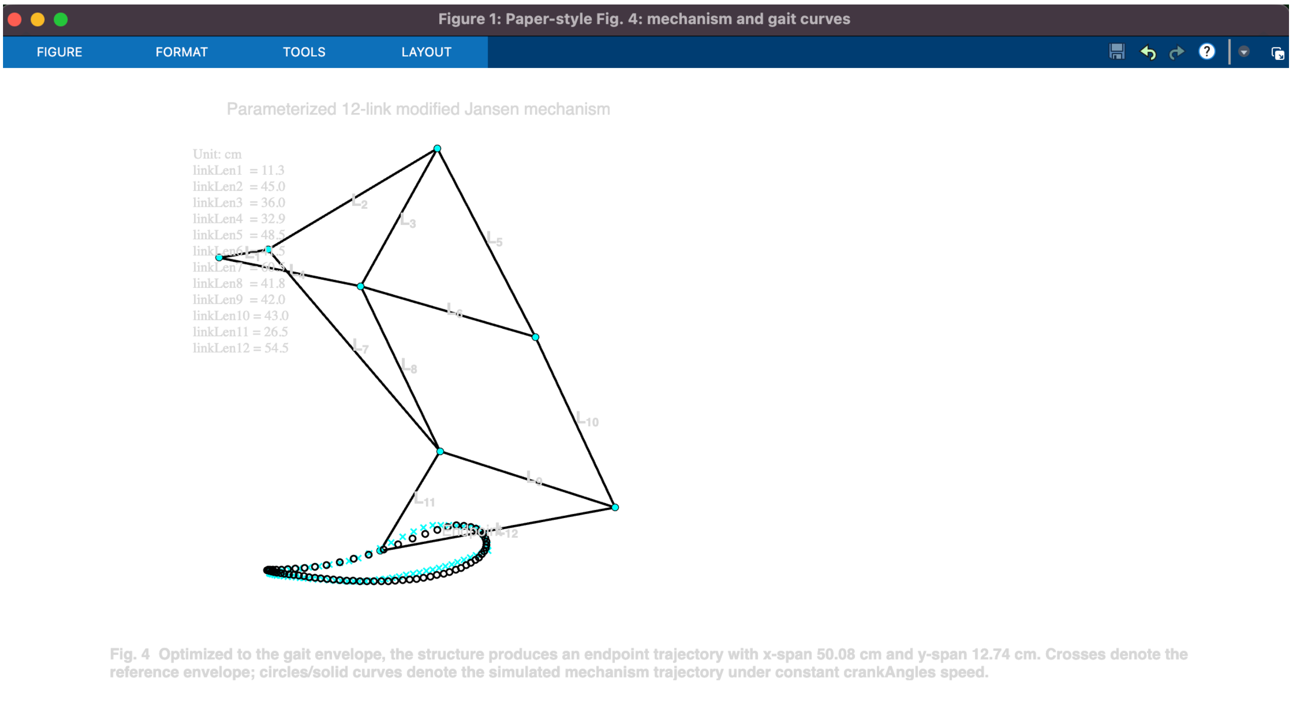

Figure 2: Parameterized 12-link modified Jansen mechanism with the simulated foot-point trajectory overlaid. Cyan crosses denote the reference gait envelope (x-span = 50.02 cm, y-span = 12.81 cm); black circles denote the simulated ptE trajectory under constant crank speed. Simulated output: x-span = 50.08 cm, y-span = 12.74 cm.

4.2 Gait Cycle Curves

When ptE's x and y coordinates are plotted against gait cycle percentage (0–100%), the horizontal (x) component shows a roughly sinusoidal forward-then-backward progression corresponding to foot swing and pushback. The vertical (y) component shows a single prominent upward peak — the foot lifting off the ground — and a flat lower region corresponding to the stance phase.

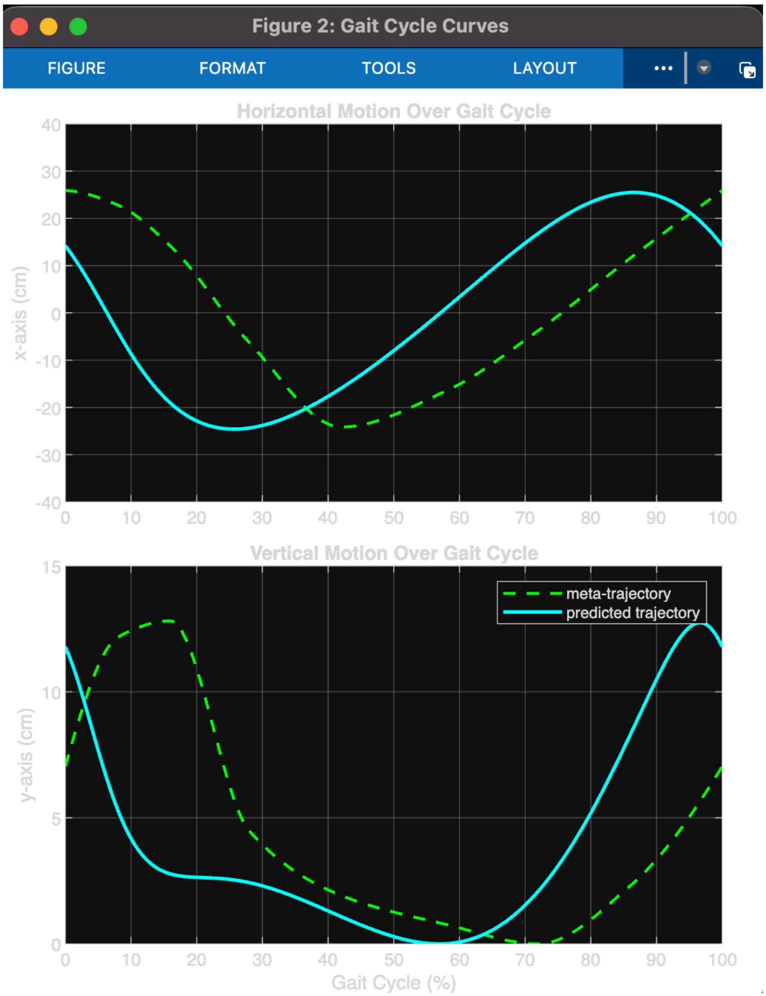

Figure 3: End-effector position over the gait cycle (0–100%). Top: horizontal (x-axis) motion — simulated trajectory (cyan) vs. meta-trajectory reference (green dashed). Bottom: vertical (y-axis) motion — showing the characteristic single lift peak of natural ankle motion.

5. Effect of Link Dimensions

Three links — L1 (crank), L4 (ground link), and L8 (a coupler link) — were identified as adjustable parameters with the most significant influence on foot trajectory shape. Their effects are summarized below:

L1 (crank length): Primarily controls the overall scale of the trajectory. Increasing L1 increases both the horizontal and vertical span proportionally. It is the most direct scalar for stride amplitude.

L4 (ground link length): Affects the horizontal span more strongly than the vertical. A longer L4 stretches the stride length without significantly altering step height, making it useful for tuning forward reach.

L8 (coupler link): Has a more complex coupled effect on both spans. Changing L8 reshapes the trajectory asymmetry — it influences the curvature of the swing phase arc and can shift the balance between stance flatness and swing height.

The three-parameter sensitivity was analyzed by independently varying each link from −4% to +4% of its nominal value across 9 samples and recording the resulting x-span and y-span. This confirms that the trajectory is sensitive to all three, but with different gain characteristics for each.

6. Comparison with Gait-Like Reference

The reference gait envelope used for comparison was reconstructed from the target span values reported in the paper: **xspan = 50.02 cm, yspan = 12.81 cm**. A smooth pchip-interpolated reference curve was generated matching these span dimensions and aligned to the simulated trajectory centroid for overlay.

The simulated foot trajectory (L1 = 11.29, L4 = 32.93, L8 = 41.78 cm) closely matches this reference envelope in both overall shape and span. The horizontal motion profile follows the reference within acceptable tolerance across the full gait cycle. The vertical profile reproduces the single-peak lift characteristic of natural ankle motion.

This agreement validates that the modified Jansen linkage with the chosen link dimensions is capable of producing a human-like ankle trajectory using only a single rotary actuator — confirming the core premise of the referenced paper.

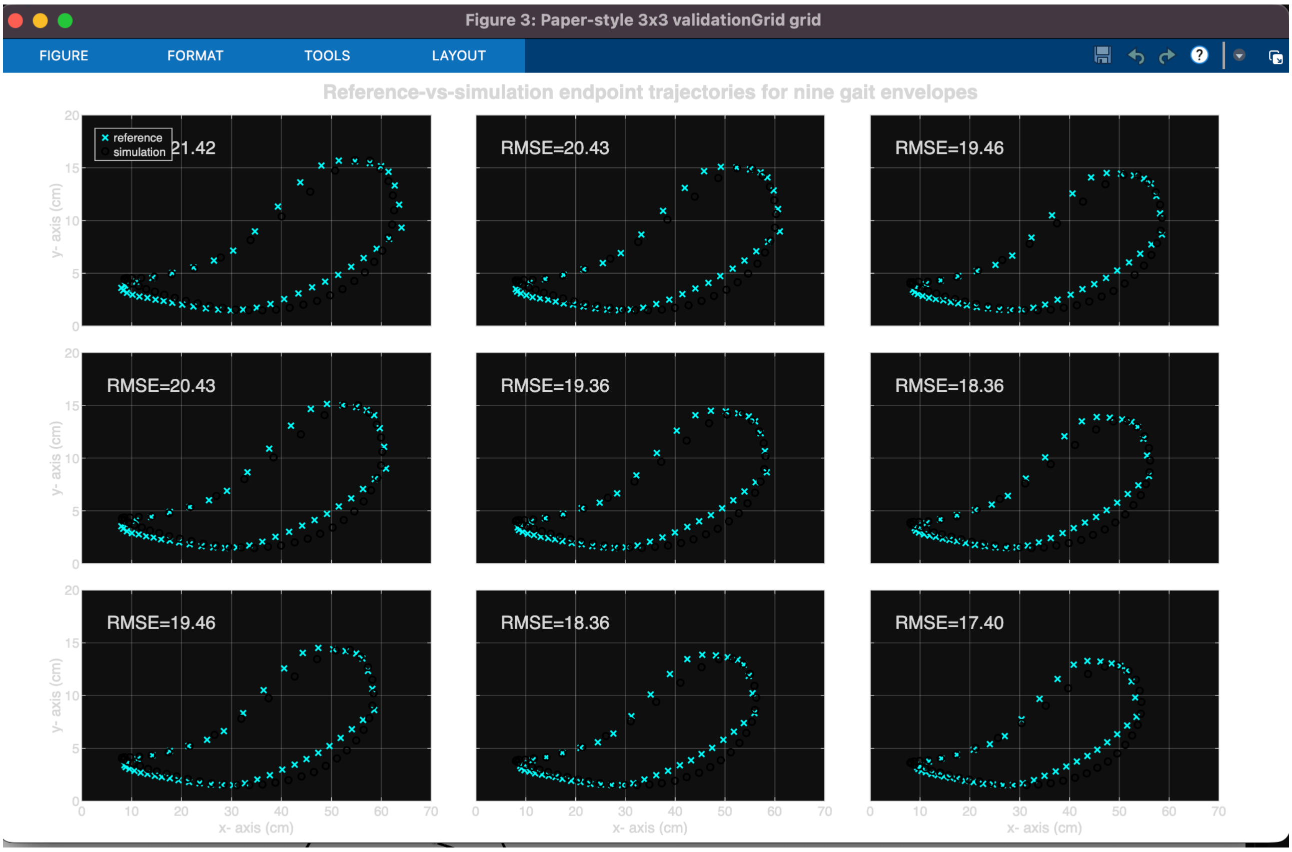

Figure 4: Paper-style 3×3 validation grid showing simulated foot trajectories (circles) against nine reference gait envelopes (cyan crosses) generated by varying L1, L4, and L8. RMSE values range from 17.40 to 21.42 cm. The bottom-right configuration (RMSE = 17.40 cm) achieves the closest match to the target gait envelope.

7. Conclusion

This project successfully simulated a 12-link, single-DOF modified Jansen gait trainer mechanism in MATLAB. The foot-point trajectory was extracted over one complete crank cycle and shown to resemble a natural gait curve, with a stride length of approximately 50 cm and step height of approximately 12 cm. The three adjustable links (L1, L4, L8) were shown to provide meaningful control over the trajectory span and shape, consistent with the findings of Shin et al. The simulation also demonstrated that a single actuator, through careful linkage design, can replicate the complex kinematics of human ankle motion, a result with clear implications for the design of passive or low-actuator walking assistive devices.

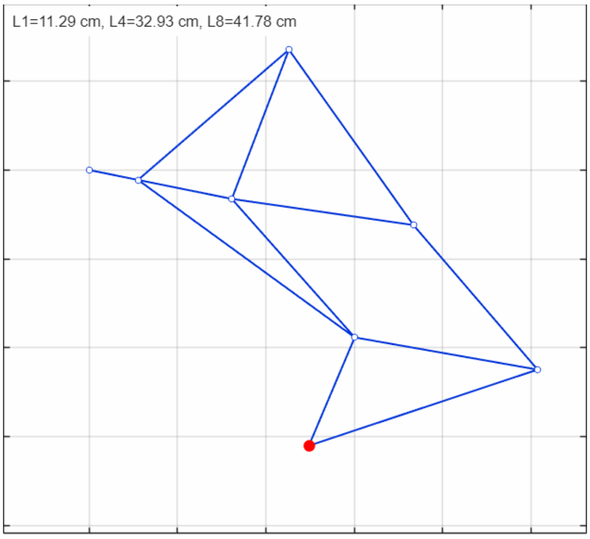

Figure 5: Animation of the simulated Jansen mechanism over one complete crank cycle. The red dot traces the foot-point (ptE) trajectory. [Full animation available in the linked GitHub repository and Google Drive.]

All simulation code, output figures, animation GIF, and the project video are available at:

- GitHub: *https://github.com/YagnikPatel22/IE410-Project-Intro.-to-Robotics*
- Google Drive: *https://drive.google.com/drive/folders/10Q-eP7eRItRPtfuS4pdbvAgAXPnDd2rc*

References

Shin, D., Deshpande, A. D., & Sulzer, J. (2018). Single-degree-of-freedom design of a Jansen-type gait trainer for human ankle rehabilitation. Journal of Mechanisms and Robotics, 10(4).

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IE410: Introduction to Robotics — Project Part B: Simulation of a Theo Jansen Walking Mechanism for Gait Generation. Course document, DAU, Winter 2026.

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